Question Paper

Exam Date & Time: 01-Jan-2026 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026
SUBJECT: CSS 2102/CSS_2102 DATA COMMUNICATION COMPUTER NETWORKS
(SPL: COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE / CYBER SECURITY / DATA SCIENCE / INFORMATION TECHNOLOGY)

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) Define data rate and bandwidth in a composite signal. (5)
Let us approximate our square wave with the waveform as shown in the equation below. Calculate the data rate and Bandwidth. Change the frequency to three times the previous frequency and calculate the data rate and the bandwidth. Explain how bandwidth affects the data rate.

$$s(t) = \frac{4}{\pi}\left[\sin(2\pi \times 2 \times 10^6 t) + \frac{1}{3}\sin(2\pi \times 6 \times 10^6 t) + \frac{1}{5}\sin(2\pi \times 10 \times 10^6 t)\right]$$

Course Outcome: "CO1" / Cognitive Level: "Application"

1B) Apply the Cyclic Redundancy Check (CRC) method to the data D = 1010001101 using the generator polynomial P = 11011. Determine the transmitted codeword after performing the CRC encoding process. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

1C) A company network consists of three departments connected as follows: Sales and HR are connected through a hub, HR and IT are connected through a switch, and the switch is connected to a router for internet access. An employee in Sales sends a file to an employee in IT. Analyse how the hub, switch, and router handle this transmission. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

2A) A company named Mindmap Solutions has been assigned the IP address 192.168.10.0/26 for its internal network. The network administrator must design subnets to efficiently allocate IP addresses to different departments. The company has the following requirements: Each department should be placed in a separate subnet. The Administration and Finance departments each need about 25 devices. The HR and Support departments each need about 10 devices. As the network engineer, you are required to design the subnetting plan based on the given IP address. Determine: (5)
i) The number of subnets created.
ii) The number of usable hosts per subnet.
iii) Network and broadcast addresses for each subnet.
iv) The subnet ranges allocated to each department.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

2B) A router receives a packet destined for 192.168.45.10. Given the forwarding table below, determine the outgoing interface using the longest-mask-match rule: (3)

Network Address	Mask Length	Outgoing Interface
192.168.0.0	/16	I1
192.168.45.0	/24	I2
192.168.45.8	/29	I3
Default		I4

Show the matching steps and justify your answer.

Course Outcome: "CO3" / Cognitive Level: "Application"

2C) Distinguish between direct and indirect delivery of packets with diagram. (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

3A) A transport layer segment of 3400 bytes is passed to the network layer for delivery from Host X to Host Y through two interconnected networks using Router R1. The first network between Host X and Router R1 allows a maximum transmission unit (MTU) of 2000 bytes, while the second network between Router R1 and Host Y allows an MTU of 600 bytes. Assume that all IP headers are 20 bytes long. If all fragments are successfully delivered to the destination, determine how the datagram is fragmented at each stage. Clearly specify the size of each fragment, data byte range, their corresponding offset values, and the flag (MF) settings for every fragment. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

3B) Analyse the actions performed by the ARP module and demonstrate how the ARP cache table would be updated to reflect the status after the given sequence of events, assuming a maximum time-out of 900 seconds and a maximum of 7 retry attempts. (3)

Table. 3(B)

State	Queue	Attempt	Time-out	Protocol Addr.	Hardware Addr.
P	5	7		229.34.48.86	
R	7		900	190.30.66.16	ACAE32457342
R	2		700	126.91.77.22	573E3242ACAD
P	9	2		19.19.76.182	
P	6	3		198.101.68.171	

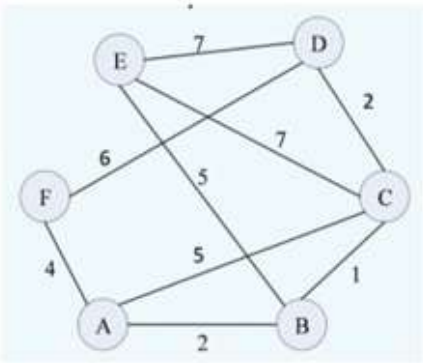
i) After 100 seconds, when Input module receives ARP reply with target IP 19.19.76.182 with Sender Physical address - 573E3242ADAD and Receiver address - 789A42242CCC.
ii) After another 100 seconds, ARP output module receives an IP datagram destined to 229.34.48.86 and ARP module retries again to send ARP request.
iii) After another 50 seconds, the ARP output module receives an IP datagram with the destination address 110.10.70.21

Course Outcome: "CO3" / Cognitive Level: "Analysis"

3C) For the given binary data 111000101010, draw the following line coding scheme. (Assume previous pulse was negative) (2)
i) Manchester
ii) Bipolar AMI

Course Outcome: "CO1" / Cognitive Level: "Application"

4A) Use Dijkstra's Algorithm to find the shortest path from vertex A to all other vertices in the following graph: (5)



Show the shortest path table and shortest path tree.

Course Outcome: "CO3" / Cognitive Level: "Application"

4B) Consider sending a large file from a host to another over a TCP connection that has no loss. (3)
i) Suppose TCP uses AIMD for its congestion control without slow start. Assuming cwnd increases by 1 MSS every time a batch of ACKs is received and assuming approximately constant round-trip times, how long does it take for cwnd increase from 6 MSS to 12 MSS (assuming no loss events)?
ii) What is the average throughput (in terms of MSS and RTT) for this connection up through time=6 RTT?

Course Outcome: "CO4" / Cognitive Level: "Application"

4C) Consider an Ethernet segment with a transmission speed of 108 bits/second and a maximum segment length of 1500 meters. If the propagation speed of the signal in the medium is 3×10^8 meters/second, what is the minimum frame size (in bits) required for collision detection? (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

5A) Compare GBN, SR, and TCP (No delayed ACK). Assume that the timeout values for all three protocols are sufficiently long such that 5 consecutive data segments and their corresponding ACKs can be received (if not lost in the channel) by the receiving host (Host B) and the sending host (Host A) respectively. Suppose Host A sends 5 data segments to Host B, and the 2nd segment (sent from A) is lost. In the end, all 5 data segments have been correctly received by Host B. (5)
a) How many segments Host A sent in total and how many ACKs has lost Host B sent in total? What are their sequence numbers? Answer this question for all three protocols.
b) If the timeout values for all three protocols are much longer than 5*RTT, then which protocol successfully delivers all five data segments in shortest time interval?

Course Outcome: "CO4" / Cognitive Level: "Analysis"

5B) A DNS resolver is trying to resolve the domain name **mail.cs.mit.edu** using an iterative query mechanism. Assume no caching. How many query-response pairs are required for complete resolution? (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

5C) The following is a dump (contents) of a UDP header in hexadecimal format. (2)
C3A20035 0024 1F2B
a) What is the source port number?
b) What is the destination port number?
c) What is the length of the data?
d) Is the packet directed from a client to a server or vice versa?

Course Outcome: "CO5" / Cognitive Level: "Remember"

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Event Name	Subject
MAKE-UP EXAM-ODD SEMESTER 2025 MIT MANIPAL	CSS 2102:DATA COMMUNICATION AND COMPUTER NETWORKS

#	Course Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	7
3	3	CO 3	18
4	4	CO 4	16
5	5	CO 5	2
		Total	**50**

* NA - CO Not Selected

Event Name	Subject
MAKE-UP EXAM-ODD SEMESTER 2025 MIT MANIPAL	CSS 2102:DATA COMMUNICATION AND COMPUTER NETWORKS

#	Level	No of Questions	Marks (%)
1	Remember	1	2(4%)
2	Understand	3	7(14%)
3	Application	7	23(46%)
4	Analysis	4	18(36%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)